FREE GUIDE

FROM 13,000 GENES TO 1 TARGET



THE TARGET DISCOVERY WORKFLOW

How to use Open Targets, SwissTargetPrediction and AlphaFold to find a druggable target for Alzheimer's.

Five steps, all free, all on a laptop. The example: finding a small-molecule target for Alzheimer's disease.

1 Pull disease genes from Open Targets

2 Filter for targets with an approved small molecule pill

3 Test a natural compound with SwissTargetPrediction

4 Check the pathway network

5 Confirm the 3D binding pocket with AlphaFold

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AISYNTH LABS

READ THIS FIRST

START WITH THE TARGET

The goal: you want to discover a new drug for Alzheimer's disease. The first step is finding a biological target.

Ask a database for every gene linked to Alzheimer's and it hands back over **13,000 results.** The real challenge is filtering that list down to a single protein you can actually drug with a small molecule pill.

1. Pull the disease genes from **Open Targets.**
2. Filter for targets that already have an **approved small molecule drug.**
3. Test a natural compound, **Huperzine A**, with **SwissTargetPrediction.**
4. Check the **pathway network.**
5. Confirm the physical **3D binding pocket** with AlphaFold.

Counts in this guide are from my pull on the day I ran it. Open Targets updates, so your count may differ slightly.

PULL THE DISEASE GENES

T **THE TOOL: Open Targets Platform**
platform.opentargets.org, free, no account needed for browsing.

A **THE ACTION**
Search "Alzheimer's disease" and open the associated targets list.

R **THE RESULT**
13,006 genes
, ranked by evidence strength. The top-ranked gene is
APP
(amyloid precursor protein).



Open Targets: Alzheimer's disease searched, 13,006 genes counted, APP at rank 1

THE SCIENCE: WHY APP IS #1, AND WHY IT IS NOT A PILL TARGET

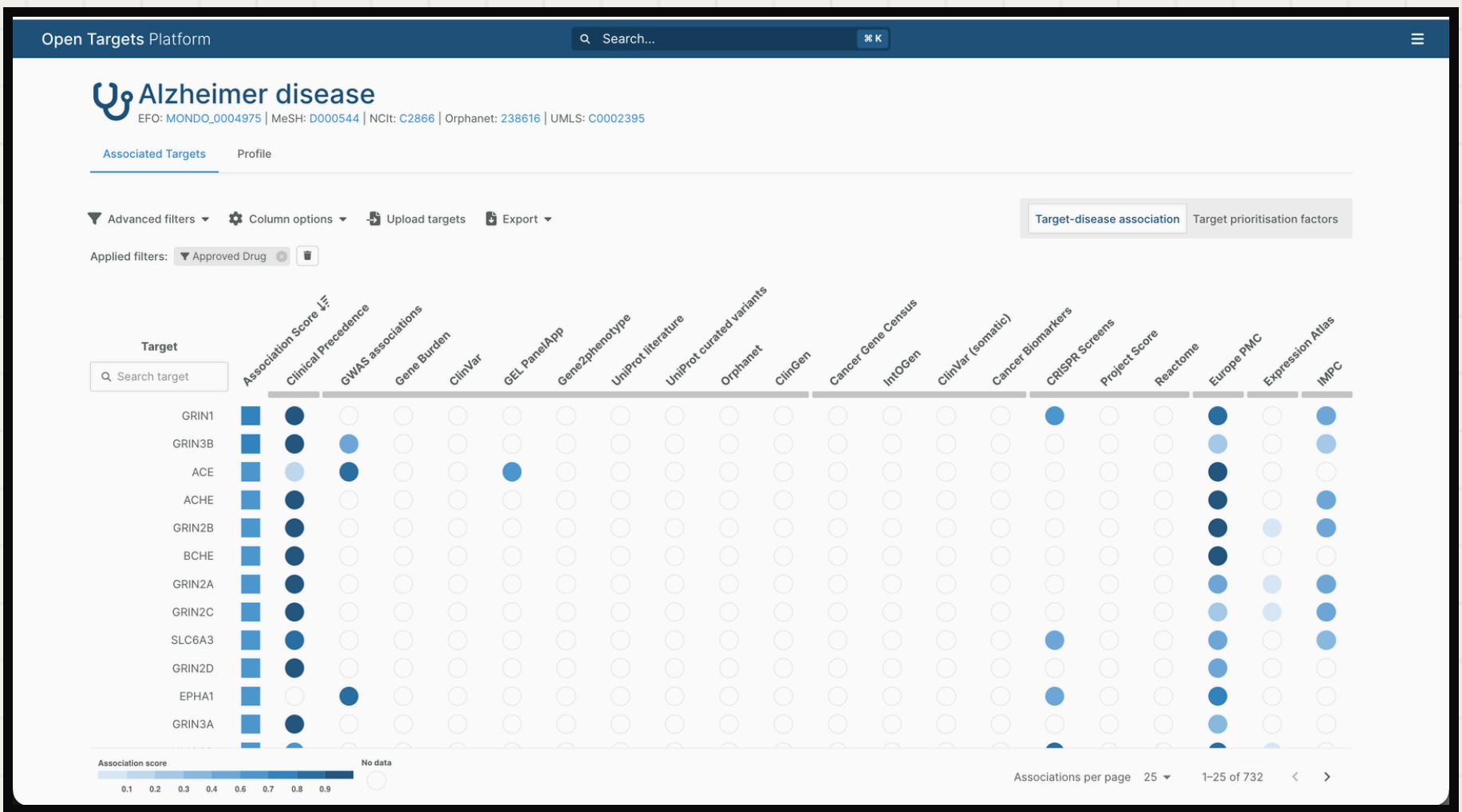
APP ranks first on research volume: it produces the amyloid that forms plaques. But APP is a massive, flexible protein with **no clear pocket** for a small molecule, and shutting it off completely is not safe. The drugs that go after its amyloid, such as **lecanemab**, are large antibodies given by **IV infusion** that cross the blood-brain barrier poorly. For a pill, APP is not the best target.

DEMAND THAT A PILL EXISTS

T **THE TOOL: Open Targets advanced filters**
Same platform, filter panel on the targets list.

A **THE ACTION**
Select tractability
"small molecule"
and
"approved drug"
. If you want a pill, only keep targets a pill already exists for.

R **THE RESULT**
The list drops from
13,006 to 732
. APP disappears.
ACHE
(acetylcholinesterase) survives, because the approved pill
donepezil
targets it.



The filter selected, and the new list: 732 genes, ACHE among them

THE SCIENCE: CHOOSE THE TARGET BY THE DRUG YOU WANT TO BUILD

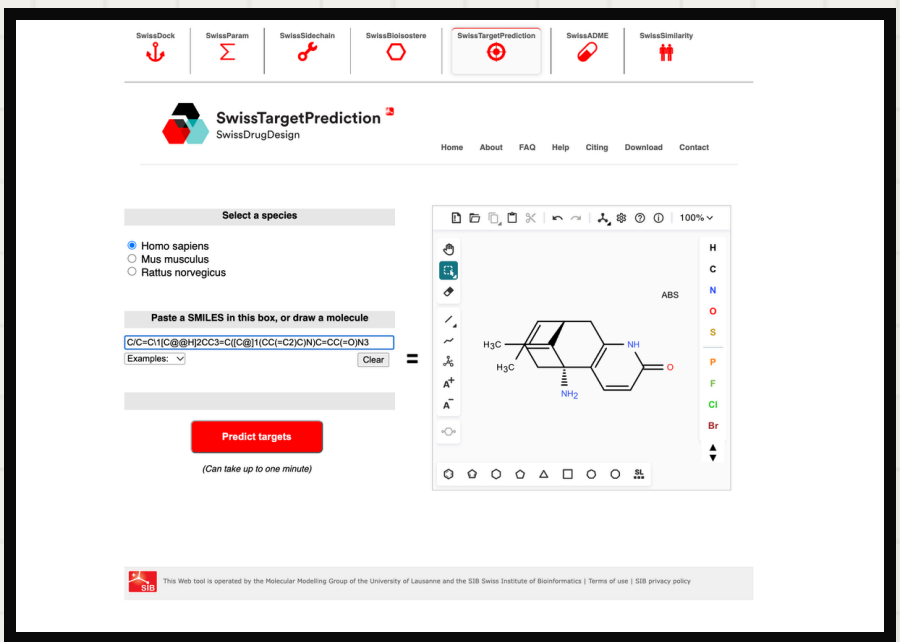
This filter proves APP has **no approved small molecule drug**, and that ACHE is **druggable in practice**: a pill already exists for it. Filter before you fall in love with a target.

TEST A COMPOUND

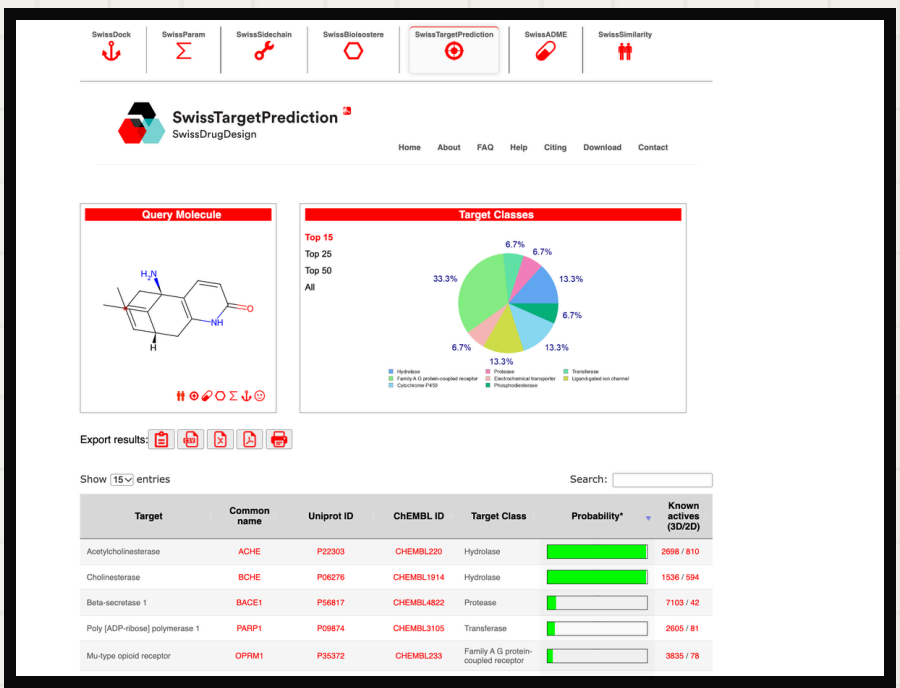
T **THE TOOL: SwissTargetPrediction**
swisstargetprediction.ch, free.

A **THE ACTION**
Take
Huperzine A
, a natural compound from Chinese club moss (*Huperzia serrata*), traditionally used for memory. Paste its SMILES or draw the structure.

R **THE RESULT**
The tool predicts Huperzine A binds strongly to
ACHE
, with a probability score.



Huperzine A in the predictor



ACHE as a top predicted target

THE SCIENCE: RATIONALE PLUS A HIT

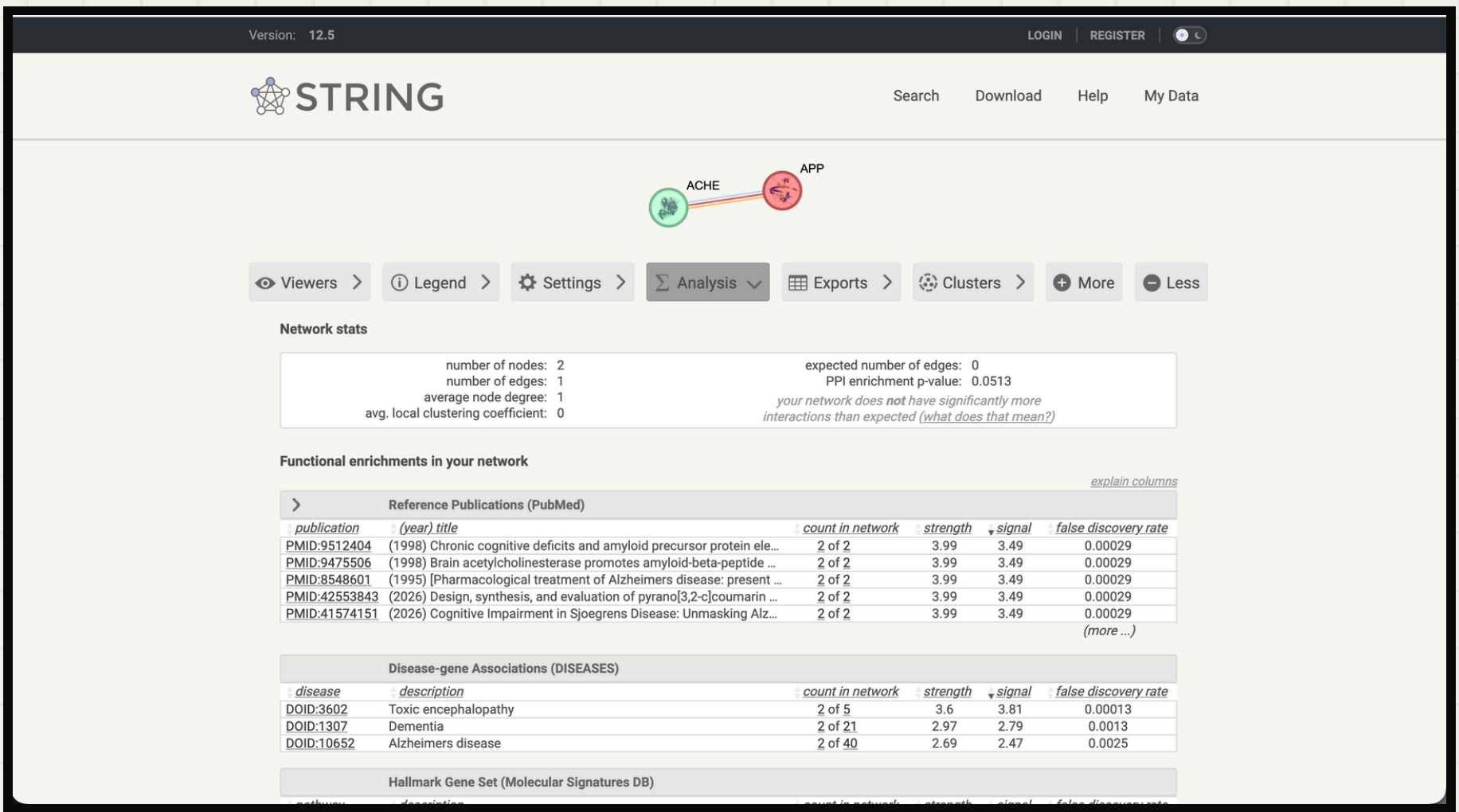
You now have both halves: a **reason to pick the compound** (traditional use for memory) and a **tool confirming it hits a protein** (ACHE) that sits on your filtered list of pill-druggable targets.

CHECK THE NETWORK

T **THE TOOL: STRING DB**
string-db.org, free.

A **THE ACTION**
Build a network of ACHE and APP and see whether they connect.

R **THE RESULT**
ACHE connects to APP. They share a biological pathway.



The network: the ACHE node connected to the APP node

THE SCIENCE: A HYPOTHESIS, NOT A PROOF

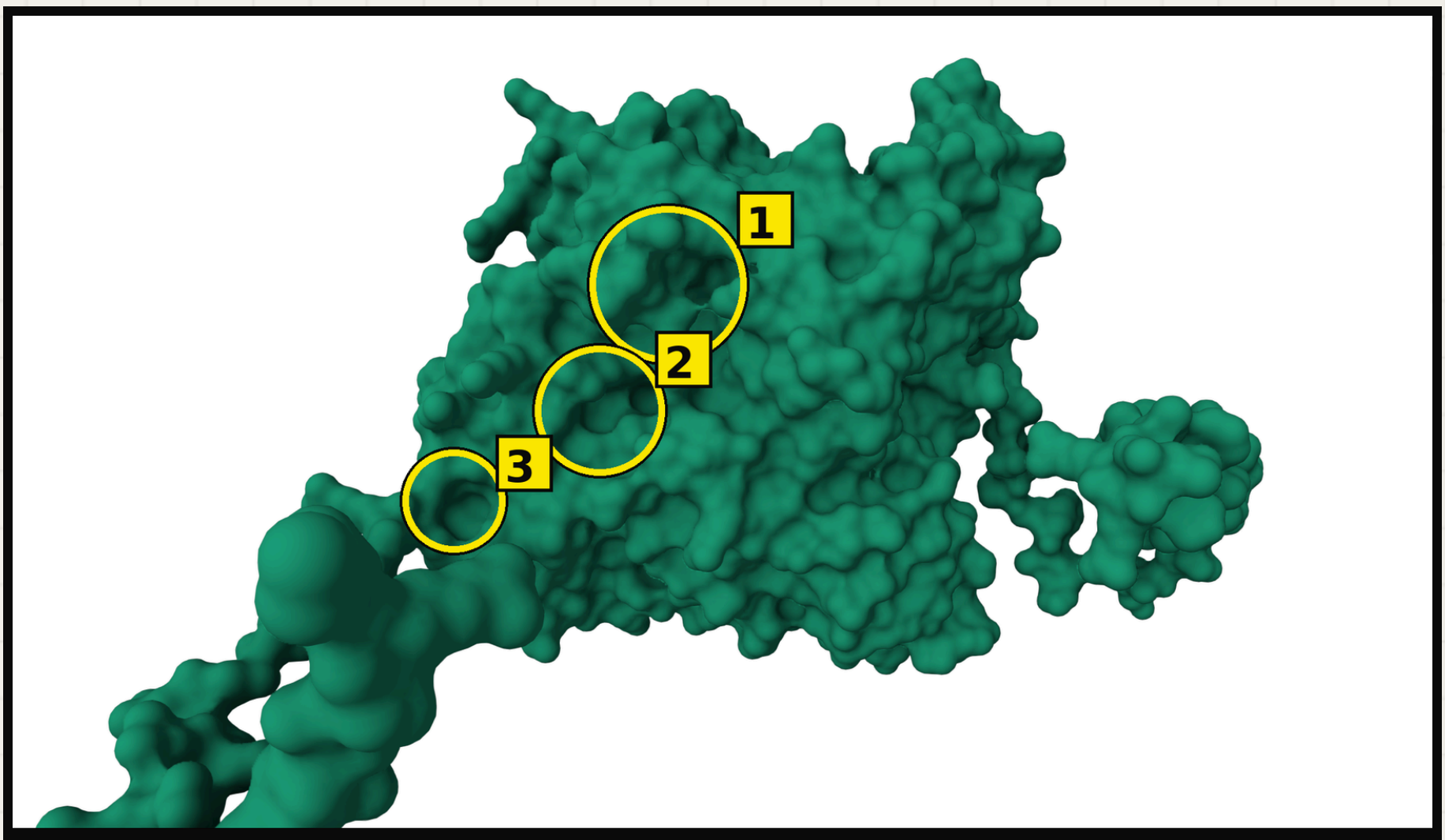
This connection means hitting ACHE **may indirectly affect the amyloid pathway driven by APP**, without drugging APP directly. Treat it as a biological hypothesis to test in the lab, not as evidence the compound works.

FIND THE PHYSICAL POCKET

T **THE TOOL: AlphaFold Database or PDB**
alphafold.ebi.ac.uk or rcsb.org, free.

A **THE ACTION**
Pull the 3D structure of ACHE and look for a binding pocket.

R **THE RESULT**
ACHE has a clear, physical pocket that a small molecule like Huperzine A can fit into.



AlphaFold model of ACHE, surface view. Circles: candidate pockets visible on the surface; #1 is the active-site gorge entrance. Confirm with a pocket detector or a bound ligand.

AlphaFold model of ACHE, surface view: candidate pockets circled, #1 is the active-site gorge

THE SCIENCE: A NETWORK EDGE IS NOT A BINDING SITE

A pathway connection only says the proteins are related. The **3D pocket** is the physical check that a drug can actually fit and bind. Both checks together make ACHE a rational target.

CONCLUSION

THE TARGET: ACHE

Your final target is **acetylcholinesterase (ACHE).**
Why:

1. It is on the filtered list: an **approved small molecule** target
2. Your compound, **Huperzine A**, is predicted to hit it
3. It connects to the main Alzheimer's pathway, **APP**
4. It has a physical **3D pocket** the drug can bind

THE LIMIT, SAID PLAINLY

This workflow gives you a **shortlist**. It does not prove Huperzine A treats Alzheimer's. What it does prove: ACHE is a rational, evidence-backed target for a small molecule. The next step is turning the molecule into computer code (SMILES) so it can be tested further.

RESOURCES

TOOLS AND NEXT STEPS

Open Targets Platform
platform.opentargets.org

SwissTargetPrediction
swisstargetprediction.ch

STRING DB
string-db.org

AlphaFold Database
alphafold.ebi.ac.uk

NEXT VIDEO

How a computer sees a molecule (SMILES and fingerprints). Comment **TARGET** on the reel and I will send you the exact links and filters I used.

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